

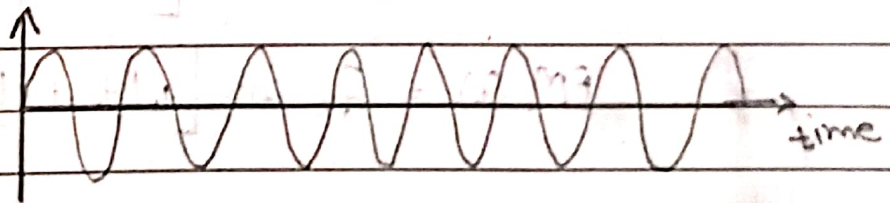
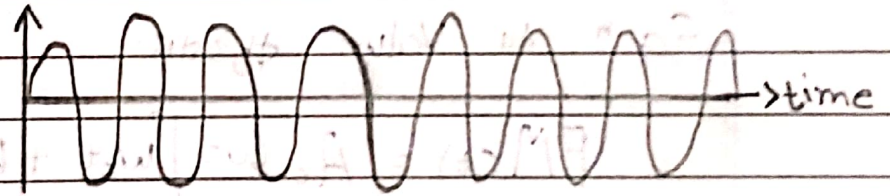
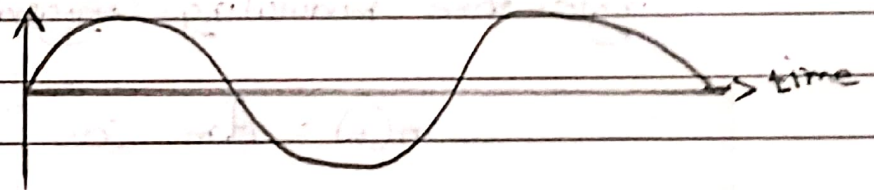
PEC Ch. 3 Assignment

Q.1 Draw and explain Frequency modulation.
Derive mathematical Equation of FM Signal.

→

Definition:- Frequency Modulation (FM) is a type of angle modulation in which the frequency of the carrier wave is varied linearly with the instantaneous frequency of the modulating signal $m(t)$. The amplitude of the carrier wave is constant.

Waveform:-



• If the message signal has a high amplitude, the frequency of the carrier wave is increased.

• If the message signal has a low amplitude, the frequency of the carrier wave is decreased.

• Mathematical Derivation :-

Unmodulated carrier wave eqⁿ :-

$$c(t) = A_c \cos(\omega_c t) = A_c \cos(2\pi f_c t)$$

General Expression for FM wave :-

$$F'M(t) = A_c \cos \left[\omega_c t + K_f \int_0^t m(t') dt' \right]$$

Single-tone modulating signal eqⁿ and eqⁿ :-

$$m(t) = A_m \cos(\omega_m t)$$

Eqⁿ in Volts eqⁿ :-

$$F'M(t) = A_c \cos \left[\omega_c t + K_f \int_0^t A_m \cos(\omega_m t') dt' \right]$$

$$F'M(t) = A_c \cos \left[\omega_c t + \frac{K_f A_m}{\omega_m} \sin(\omega_m t) \right]$$

Final Single-tone FM wave :-

$$F'M(t) = A_c \cos \left[\omega_c t + m_f \sin(\omega_m t) \right]$$

Q.2 Find modulating index of FM signal.

Definition :- FM Signal or Modulation Index (m_f) is maximum frequency deviation (Δf) and maximum modulating frequency (f_m) ratio.

Formula :-

$$m_f = \frac{\Delta f}{f_m}$$

$$\text{or } m_f = \frac{K_f A_m}{f_m}$$

m_f = Modulating index of FM.

Δf = maximum frequency deviation in Hz.

f_m = Modulating Signal Frequency in Hz.

K_f = Frequency Sensitivity / Frequency deviation Constant

A_m = Amplitude of the modulating signal.

Q.3 Compare Narrowband FM and Wideband FM.

S. No.	Parameter	Wideband FM	Narrowband FM
1.	Modulation index	Greater than 1	less than or slightly greater than 1
2.	Maximum deviation	75 kHz	5 kHz
3.	Range of mod. frequency	30 Hz - 15 kHz	30 Hz to 3 kHz
4.	Maximum modulation index	5 - 2500	Slightly greater than 1
5.	Bandwidth	large	Small
6.	Application	Entertainment broadcasting	Mobile Comm.

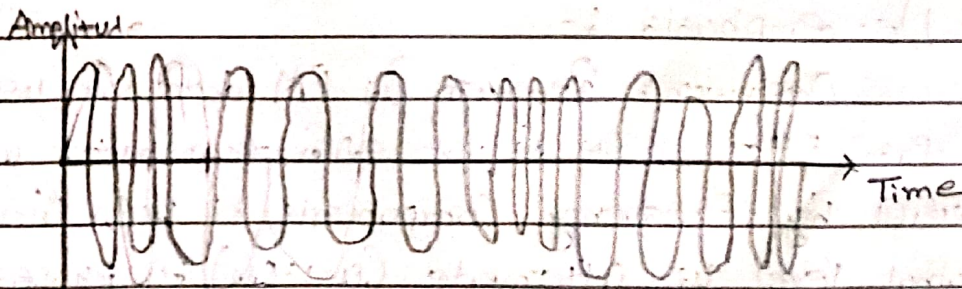
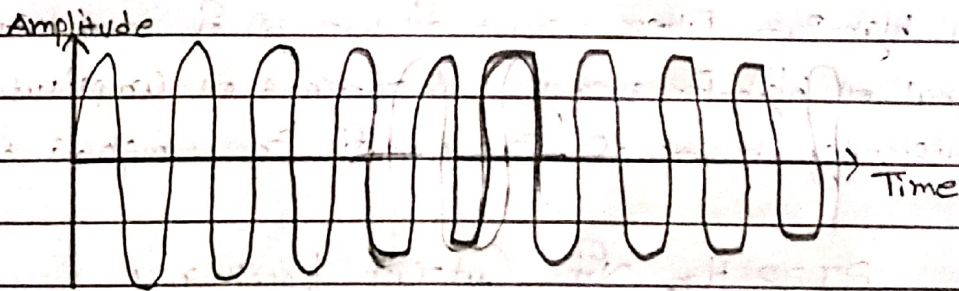
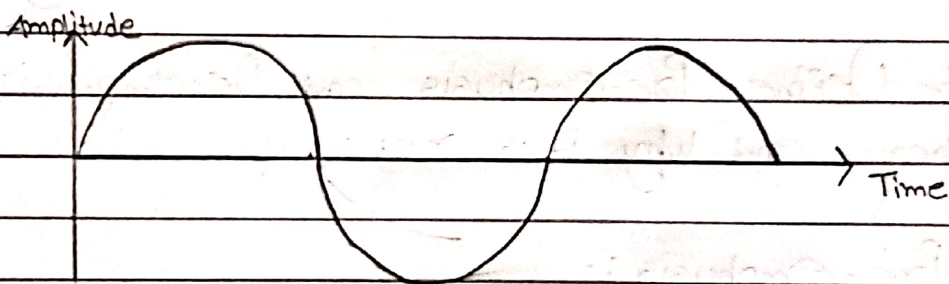
Q.54 What is Phase Modulation? Draw Waveform and Write its application.

→ Definition:- Phase Modulation (P.M) or angle modulation is a technique of PM carrier wave in instantaneous phase angle modulating message signal or amplitude change but its carrier of amplitude is Frequency Constant of

Eqⁿ :-

$$v = A \sin(\omega_c t + m_p \sin \omega_m t)$$

Waveforms :-



Amplifier

Applications of PM:-

- Digital data transmission all use PM & all use Phase-shift & PSK, BPSK, QPSK & others.
- Wireless technologies like WiFi, GSM, and Satellite TV transmission all data signal.
- FM broadcasting all Stereo and Auxiliary signals encode & decode.
- RFID and NFC Communication Systems all.

Q.5 Define Pre-emphasis and De-emphasis, Where and Why it is required?

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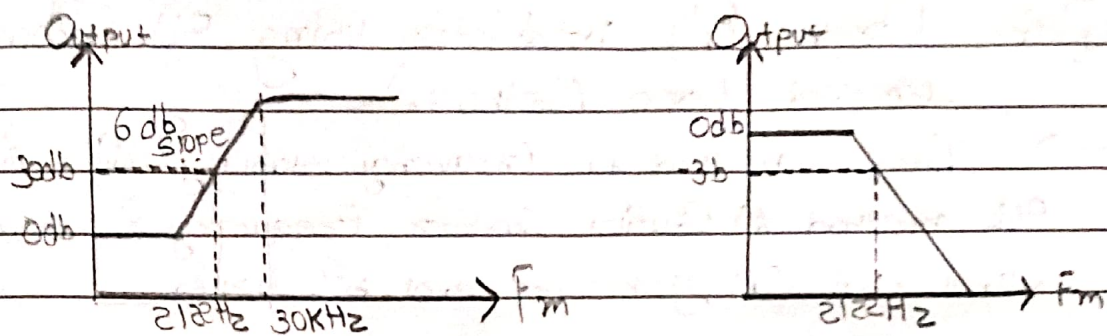
• Pre-emphasis:-

Definition: Transmitter or FM modulator use a high-pass filter circuit to use all the modulating signal at high-frequency components at amplitude is artificially boost in process of Pre-emphasis & &.

2122 Hz is Cut-off frequency &.

• De-emphasis:-

Definition: Receiver or FM detector use a low-pass filter circuit to use all the transmitter use original high-frequency components at level signal original level is attenuate (reduce) in process of De-emphasis & &.



• Pre-emphasis

• De-emphasis

• Where and Why it is required?

→ Pre-emphasis transmitter section & modulator
 હેઠળ અને De-emphasis receiver section & demodulator
 હેઠળ હોવા છે.

→ કોઈ શું ?

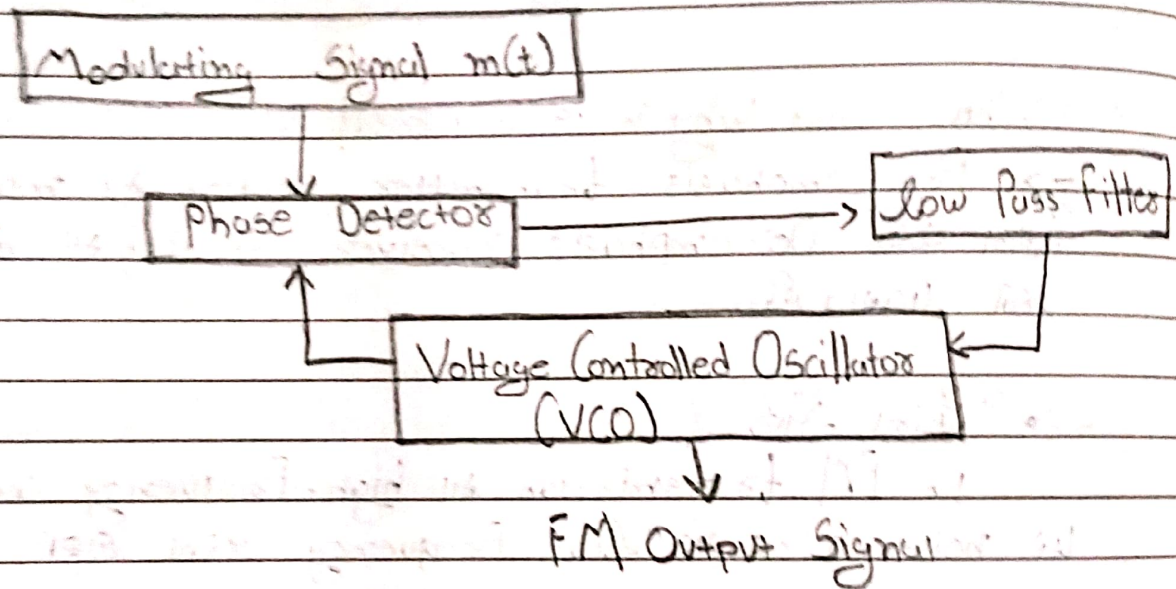
1. FM transmission માં high-Frequency audio component
 નો noise નો અસર low frequency કરતાં ઓછી હોય છે.

2. જો Pre-emphasis અને de-emphasis નો ઉપયોગ નો
 high frequencies નો signal to noise ratio ઘટાડવા
 માટે થાય છે.

3. Pre-emphasis high frequencies નો effective
 modulation index ઘટાડે છે. SNR improve થાય છે, અને
 de-emphasis original balanced audio restore થાય છે.

Q.6 Draw FM modulator using PLL (Phase Locked Loop Method.)

→ Direct method માં Frequency stability બાંહેધી મળતી હોવાથી PLL method ની ઉચ્ચતી Carrier Frequency ની stable ચાલવા વાળું FM generate કરી શકે છે.



Explanation:

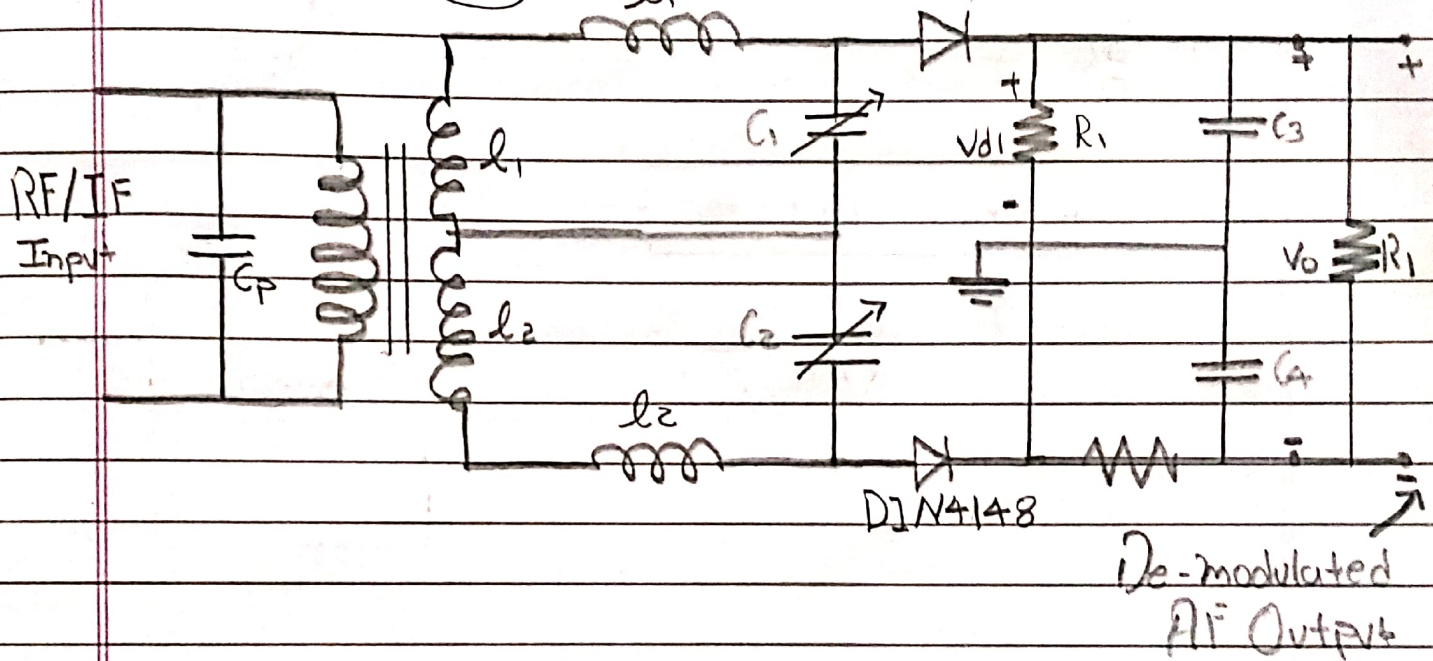
• Modulating message signal ની VCO ના ફ્રીક્વેન્સી ઉપર પર અથવા Phase detector ના ફેરફાર માંથી આપણને VCO ની આઉટપુટ ફ્રીક્વેન્સી સીધું મેળવવા મુશ્કેલી પડે છે.

• PLL લૂપ કોન્ટ્રોલમાં પરબંધ રાખેલું છે જેના કારણે થતી ત્રુટી વધી અંકુશ કરેલા FM સિગ્નલ મળે છે.

Q.7 Draw and Explain FM demodulator: Balanced Slope Detector.

→

Circuit Diagram:-



Explanation:-

- Balanced slope detector is a slope detector back-to-back connect. It is a signal single slope detector. It has linearity and efficiency.

- Center-tapped transformer in upper secondary circuit carrier frequency is not tune, and lower secondary circuit carrier frequency is not tune.

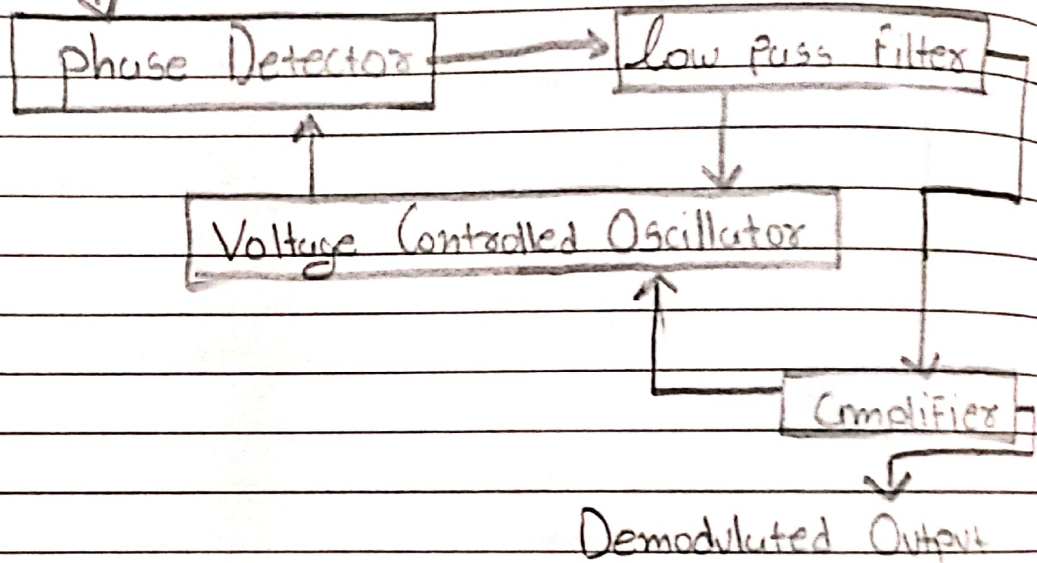
- Incoming FM signal is split by two diode detectors D1 and D2. The output of D1 and D2 is 180° out-of-phase signal.

- Final output is the difference of the output of the two detectors. It is a linear S-curve. The original message signal is recovered.

Q. 8 Draw & explain PLL (Phase locked loop) demodulator.

Block Diagram:-

FM Input



Working Principle:-

• No Modulation :- When input is an unmodulated carrier, the incoming carrier center frequency is locked to the VCO at its tune or voltage mid-position. It is set to zero.

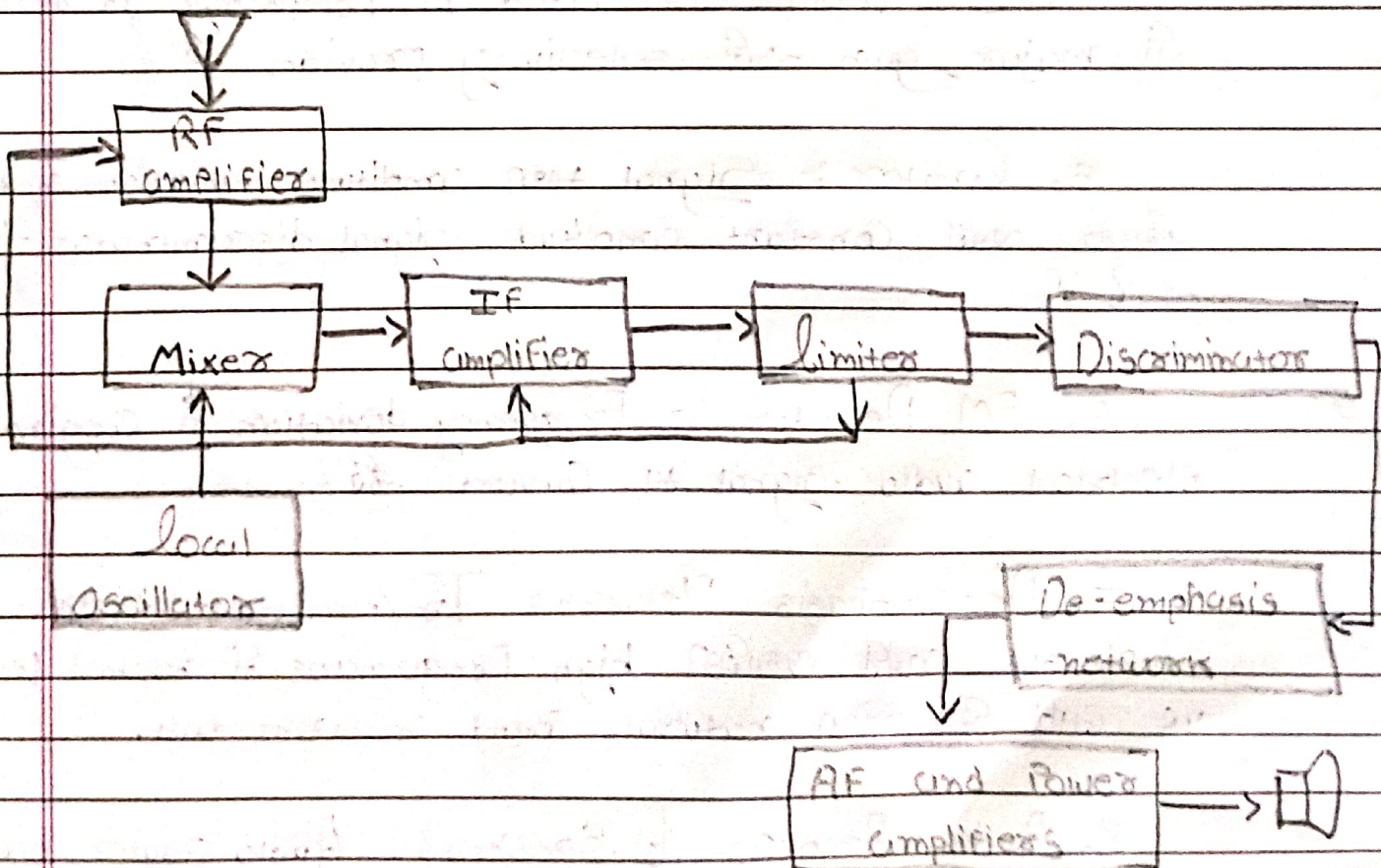
• Frequency Deviation :- When incoming carrier frequency deviates, the loop tries to lock and correct the frequency.

• Tracking :- Loop lock remains and VCO frequency follows the input signal. When the input frequency changes, the VCO frequency also changes to track it.

• Output Recovery :- True line voltage is variation incoming signal and modulation it exact follow.

• Amplification :- The voltage variations in buffer amplifier will amplify given original demodulated audio signal and send.

Q.1 Draw and Explain block diagram of FM receiver.



• Working of each block:-

1. Antenna :- Space mein FM radio waves capture kr kr weak electrical signals generate kr kr

2. RF Amplifier :- Weak signal ko amplify kr kr, unwanted noise filter kr kr kr receiver ki sensitivity badh kr.

3. Mixer & Local Oscillator :- Heterodyning process mein high RF signal ko fixed intermediate frequency ki convert kr kr.

4. IF Amplifier :- Fixed IF frequency ki receiver ki major gain kr kr selectivity provide kr kr.

5. Limiter :- Signal ki amplitude variations remove kr kr kr constant amplitude signal discriminator ki badh kr.

6. FM Detector :- Frequency deviation ki original electrical audio signal ki convert kr kr.

7. De-emphasis Network :- Transmitter ki pre-emphasis ki kr kr high frequencies ki normal level ki badh kr kr kr natural sound restore kr.

8. Audio Amplifier & Speaker :- Audio power boost kr kr loudspeaker drive kr kr kr original sound badh kr.

Q.10 Compare AM, FM & PM
→

Parameter	Amplitude Modulation	Frequency Modulation	Phase Modulation
Definition	Carrier of Amplitude message from Change ΔA & $\Delta \phi$.	Carrier of Frequency message from Change Δf & $\Delta \phi$.	Carrier of Phase message from Change $\Delta \phi$ & Δf .
Equation	$s(t) = [A_c + m(t)] \sin \omega_c t$	$s(t) = A_c \sin[\omega_c t + m_f \sin \omega_m t]$	$s(t) = A_c \sin[\omega_c t + m_p \sin \omega_m t]$
Constant Parameter	Frequency and Phase Constant	Amplitude and Phase Constant	Amplitude and Frequency Constant.
Frequency Range	535 kHz - 1705 kHz	88 MHz - 108 MHz	High frequency bands
Bandwidth	Small	Large	Moderate
Noise Immunity	Poor	Excellent	Better than AM but inferior to FM.
Circuit Complexity	Very Simple Circuits	Complicated Circuits	Complicated Circuitry
Modulation Index	$m = A_m / A_c$	$m_f = \Delta f / f_m$	$m_p = \Delta \phi \propto A_m$